SANTOSH KANUMURI

Programmer Analyst | Full Stack & AI Engineer

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, SQL, Bash, NodeJS

Frameworks: FastAPI, Django, Flask, ReactJS, NextJS, LangChain, Pydantic, Streamlit

AI / ML: LLMs (OpenAI, Llama, Ollama), RAG, MCP (Model Context Protocol), AI Agents, OCR, NLP, Computer Vision, Vector Databases (Pinecone, Chroma)

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, ECS, Fargate), Docker, Podman, Kubernetes, Linux, CI/CD, Cron Jobs

Databases: PostgreSQL, MongoDB, DynamoDB, Amazon Aurora, Pinecone

Tools & Standards: Git, ServiceNow, REST APIs, Postman, Agile/Scrum, WCAG 2.1, VPAT, JIRA, Selenium, MinIO

EDUCATION

Texas Tech University — Aug 2023 – May 2025

Master of Science in Computer Science | GPA: 3.85 / 4.00 | Parallel Processing · Computer Security · Advanced Operating Systems · Neural Networks

WORK EXPERIENCE

Texas Tech University – Enterprise Applications — May 2025 – Present

- *Programmer Analyst III (AI Engineer)*
  - **Technical Lead & AI Application Architect:** Serve as technical lead and primary contributor for the full-stack design, development, testing, and production deployment of enterprise-grade AI applications owning the complete SDLC from requirements gathering and feasibility analysis through implementation, QA, and documentation; regularly consult stakeholders to translate user requests into programmable requirements and advise on critical technology decisions aligned with university policy and standards.
  - **ASK_RAIDER – Conversational AI Platform:** Engineered ASK RAIDER, a university-wide conversational AI platform built on Retrieval-Augmented Generation (RAG) architecture with LLM inference via MCP-based data integrations, covering operating policies, system regulations, dining services, and a structured IT knowledge base enabling autonomous resolution of Tier-1 support queries and projecting a 40% reduction in IT Help Central ticket volume, with intelligent escalation routing reserved for complex issues requiring human intervention.
  - **Operational Monitoring & Visualization:** Developed a real-time campus operational monitoring and visualization application by integrating live data feeds directly from ServiceNow ITSM, surfacing the live status of classroom AV systems, digital signage, networked printers, and dining facilities on an interactive map - enabling facilities, IT, and operations teams to locate the nearest functional resource instantly and reduce mean time to resolution for equipment and service issues.
  - **AI-Driven Accessibility & VPAT Compliance:** Architected a full-suite web accessibility compliance platform enabling development teams to run automated WCAG 2.1 audits across university web properties, aggregate findings in a centralized dashboard, and generate VPAT reports using LLM agents with in-house model inference for secure, privacy-preserving testing - standardizing remediation workflows and regulatory reporting across 20+ digital properties while eliminating reliance on external accessibility audit vendors.
  - **PDF Remediation Pipeline – 90% Cost Reduction:** Built a production-grade PDF remediation service leveraging large language models and OCR models (supporting both native digital and scanned documents) with automated batch-processing pipelines, a self-service web portal, and full audit logging - replacing external vendor dependencies (Adobe Acrobat Services, AWS) with an in-house solution that reduced per-document remediation cost by 90% while increasing throughput and maintaining regulatory compliance.
  - **Legislative Compliance & Workflow Automation:** Delivered a LLM-powered syllabus compliance verification system that programmatically validates course syllabi against state legislative requirements and university academic policy, auto-flags non-compliant content with actionable remediation guidance, and routes flagged documents through a structured approval pipeline - improving regulatory accuracy, reducing manual legal review overhead, and streamlining centralized syllabus management for the registrar's office.

    Stack: Python, FastAPI, NextJS, LangChain, RAG, MCP, LLMs (OpenAI / Llama / Ollama), OCR, Docker, Kubernetes, Helm, ServiceNow API, AWS, PostgreSQL, WCAG 2.1, AI Agents, REST APIs

Texas Tech University — Sep 2023 – May 2025

- *Graduate Assistant – Full Stack Developer*
  - **Computer Vision & Vector Search:** Engineered a computer vision–powered art similarity search engine processing 35,000+ artworks using Pinecone vector embeddings and deep learning models, achieving 92% price-trend prediction accuracy and

70% faster retrieval; built and maintained automated ETL pipelines ingesting 1,000+ new artworks monthly from 3 auction house sources with cloud-based scheduling and email alerting on pipeline failures.

- **AI Search Agent & Cloud Deployment:** Developed an intelligent multi-source web search agent integrating Google and DuckDuckGo APIs with GPT-powered result summarization and citation generation, achieving 40% higher information accuracy than single-source queries; containerized the service with Docker, deployed on AWS ECS, and implemented a CSV batch mode processing 500+ data points per hour for high-throughput research workflows.
- **Local LLM & NLP Classification System:** Architected a local LLM (Llama 3.2) innovation detection and classification system that scraped and analyzed 1,000+ company websites, classified research domains with 88% accuracy using NLP and zero-shot prompting, and mapped innovation spectrums across sectors reducing manual classification time by 85% and enabling cross-sector trend identification for business intelligence reporting.
- **NLP Analytics & Data Warehousing:** Implemented a personality and talent analytics system using Google Gemini API to process 900+ LinkedIn profiles with NLP-based trait extraction, reducing hiring mismatches by 35% and shortening recruitment cycles by 2 weeks; consolidated fragmented financial data from Pitchbook across 5+ sources and 30,000+ records into a centralized data warehouse, producing 45% faster market intelligence reports.

Stack: Python, FastAPI, Flask, MongoDB, DynamoDB, AWS (S3, EC2, ECS, Lambda, Step Functions), Docker, PyTorch, NLP, Computer Vision, Pinecone, OpenAI, Gemini API, ETL

PricewaterhouseCoopers India — July 2022 – July 2023

- *Full Stack Developer*
  - **Enterprise Full-Stack Development:** Led full-stack development of CitiBank's internal loan document tracking system as part of a 30-person cross-functional team - engineered Django microservices backed by Amazon Aurora (PostgreSQL) processing 3,000+ daily document requests at 99.8% uptime, cutting document retrieval time by 65% and improving loan processing efficiency by 40%.
  - **Containerization & Frontend Engineering:** Containerized all application components with Docker for environment parity across development, testing, and production, reducing deployment errors by 85%; developed ReactJS front-end with granular role-based access control for concurrent users, decreasing onboarding and user training time by 3 weeks.
  - **Stakeholder Collaboration & Project Management:** Built a centralized project management platform integrating task tracking, sprint assignment, and client feedback workflows; collaborated with stakeholders through extended dialogue to translate business requirements into technical specifications and ensured all software protocols remained compliant with organizational standards.

Stack: Python, Django, ReactJS, REST APIs, PostgreSQL, Docker, AWS (EC2, S3, Lambda, Aurora), Git, Postman, Agile/Scrum

Systema Automation — July 2021 – June 2022

- *Python Developer Intern*
  - **Dashboard Development & Team Lead:** Designed and implemented a real-time metrics visualization dashboard using Django and Chart.js, processing 5,000+ daily data points and reducing analysis time by 65%; led a 3-developer Agile team delivering all 12 sprint milestones on schedule with 94% automated test coverage.
  - **API Integration & Quality Assurance:** Integrated 5+ third-party REST APIs for real-time data collection and automated data validation processes that corrected 98% of input errors pre-insertion, reducing API latency by 70% and server load by 45%; engineered a responsive mobile-first UI increasing mobile engagement by 40%.

Stack: Python, Django, REST APIs, JavaScript, HTML5, CSS3, Chart.js, PostgreSQL, Git, Agile

SELECTED PROJECTS

ASK_RAIDER – TTU Institutional AI Chatbot — *Feb 2025*

Production RAG-based chatbot prototype (LangChain + Chroma + LLM + MCP integrations) delivering semantic retrieval across TTU institutional documents; foundation for the university's enterprise ASK_RAIDER platform serving operating policies, IT knowledge base, and student services.

Alzheimer's MRI Classification – Zero-Shot VLM — *Nov 2024*

Vision-language model system using the ZLaP (Label Propagation Zero-Shot) technique to classify Alzheimer's MRI stages - non-demented, very mild, mild, and moderate - without task-specific labeled training data, demonstrating zero-shot transfer learning for medical imaging.

CourseSync – Real-Time University Course Tracker — *Sep 2024*

Full-stack application (React + Flask) with automated web scraping pipelines and real-time notification system for course availability monitoring. Containerized backend deployed on Heroku; React frontend on Netlify with live data refresh and email/browser push alerts.

CERTIFICATIONS & AWARDS

AWS – Certified Cloud Architect · **MongoDB – Certified Associate Developer** · **Google – Cybersecurity Professional Certificate**

Digital Acumen Award – Awarded by PricewaterhouseCoopers India for exceptional technical contribution and client impact.